

1. What is SAP?

SAP (Systems, Applications, and Products in Data Processing) is an enterprise software platform used by organizations to manage and integrate core business processes such as finance, human resources, procurement, manufacturing, supply chain, and sales within a single centralized system.

Instead of separate disconnected tools, SAP provides a unified environment where all departments share real-time data, improving operational efficiency, accuracy, and decision-making across the organization. It is primarily an ERP (Enterprise Resource Planning) system designed to handle large-scale transactional workloads for medium to large enterprises.

2. What is ABAP?

ABAP (Advanced Business Application Programming) is a high-level programming language developed by SAP SE for building and customizing applications within the SAP ecosystem.

It is used to create reports, interfaces, forms, data conversions, enhancements, and workflows that support enterprise business processes. ABAP is tightly integrated with SAP's database and application layer, allowing developers to process large volumes of transactional data reliably while maintaining system stability.

It is primarily an enterprise application language, optimized for scalability, security, and business logic rather than general-purpose software development.

3. What are the different types of ABAP programs?

The main types of ABAP programs are:

- **Report Programs:** Executable programs used to read data from database tables, process it, and display output (often as lists or ALV reports).
- **Module Pool Programs (Dialog Programs):** Used to build interactive applications with screens, user inputs, buttons, and navigation. Common in transactional workflows.
- **Function Groups:** Containers for function modules that encapsulate reusable business logic and can be called from different programs.
- **Interface Programs:** Used for data exchange between SAP and external systems (for example via IDocs, BAPIs, RFC).
- **Forms (SAPscript / Smart Forms / Adobe Forms):** Programs designed for formatted document generation such as invoices, purchase orders, and delivery notes.
- **Enhancements / Exits:** Custom logic added to standard SAP behaviour without modifying the original code, allowing system customization while remaining upgrade-safe.

4. What is the SAP Data Dictionary?

The SAP Data Dictionary (DDIC) is a centralized repository in the SAP SE system that stores and manages metadata — meaning the technical definitions of data used across the SAP environment.

It defines objects such as:

- **Database Tables:** Database tables are physical structures that store application data in rows and columns. They form the foundation for data storage and retrieval in SAP systems.
- **Views:** Views are virtual tables created from one or more database tables without storing data themselves. They simplify data access and present information in a specific logical format.
- **Indexes:** Indexes are performance optimization tools that speed up data retrieval from database tables. They reduce the time required to search for records by creating efficient lookup paths.
- **Data Elements:** Data elements define the semantic meaning of a field, such as its label and description. They act as reusable metadata that ensures consistency across tables and programs.
- **Domains:** Domains define the technical properties of a field, including data type, length, and valid value ranges. They enforce data integrity by restricting what can be entered.
- **Structures:** Structures are data objects that group multiple fields together without storing data in the database. They are mainly used for temporary data handling and program logic.
- **Search Helps:** Search helps provide possible input values to users through an F4 help feature. They improve accuracy and usability during data entry.
- **Lock Objects:** Lock objects are used to prevent multiple users from modifying the same data simultaneously. They help maintain data consistency and avoid transaction conflicts.

The Data Dictionary ensures data consistency, integrity, and reuse by allowing these definitions to be created once and referenced throughout multiple programs and modules. It also enables centralized control over how data is stored, validated, and accessed within the system.

5. What are transaction codes in SAP?

SAP transaction codes (T-codes) are short alphanumeric shortcuts that directly execute specific tasks, programs, or menus, bypassing complex menu navigation. They are essential for efficiency across SAP modules (e.g., VA01 for sales orders, ME21N for purchase orders), enabling quick access to screens like SE38 for ABAP or BP for business partners.

6. Important t-codes in SAP.

Category	T-Code	Purpose
ABAP Development	SE38	Create, edit, and execute ABAP reports
ABAP Development	SA38	Execute existing programs directly
ABAP Development	SE80	Object Navigator for full development (programs, classes, FMs)
Function Modules	SE37	Create, test, and manage function modules
OOABAP	SE24	Class Builder for object-oriented development
Data Dictionary	SE11	Create/manage tables, views, domains, data elements, structures, lock objects
Data Dictionary	SE12	Display dictionary objects (read-only)
Table Data	SE16	Display table contents
Table Data	SE16N	Advanced table display with better usability
Database Utility	SE14	Adjust database objects and manage table activation
Lock Monitoring	SM12	View and manage active data locks
Transport Requests	SE09	Workbench transport organizer
Transport Requests	SE10	Customizing transport organizer
Debugging	/H	Activate debugger while running a program
Performance	SE30	Runtime analysis for performance checks
Performance	SAT	Advanced runtime analysis tool
Code Quality	SCI	Run Code Inspector for static code checks
Smart Forms	SMARTFORMS	Design and maintain smart forms
BADI	SE18	Define Business Add-Ins
BADI	SE19	Implement Business Add-Ins
Enhancements	SMOD	Find customer exits
Enhancements	CMOD	Implement customer exits
System Errors	ST22	Analyze runtime dumps
Authorization	SU53	Check authorization failures
Background Jobs	SM37	Monitor scheduled/background jobs
SQL Trace	ST05	Trace database queries for performance analysis

7. What is R/3 Architecture in SAP?

SAP R/3 Architecture is a three-tier client-server architecture designed to separate responsibilities for better scalability, performance, and system management.

- **Presentation Layer:** This is the user interface where end users interact with SAP through SAP GUI or web interfaces. It handles input and displays output but does not process business logic.
- **Application Layer:** The application server processes business logic, executes ABAP programs, and manages communication between the presentation and database layers. It is the core processing engine of the SAP system.
- **Database Layer:** This layer stores all system data, including business transactions and configuration information. It ensures data consistency, security, and efficient retrieval.

8. What is BDC programming?

BDC (Batch Data Communication) is a technique in SAP SE systems used to transfer large volumes of data into SAP by automatically executing standard transactions.

It works by simulating user input — entering data into screens and fields exactly as a user would — but in a controlled, automated batch process.

Key characteristics:

- Used for mass data migration or uploads
- Follows existing transaction screens
- Does not require creating new database logic
- Helps maintain data integrity by leveraging standard validations

Two Main Methods:

- Session Method: Data is processed in batches; safer and suitable for large datasets.
- Call Transaction Method: Executes transactions immediately; faster but less robust for error handling.

9. What is a transparent table?

A transparent table is a database table in SAP where the table definition in the Data Dictionary matches exactly with a single physical table in the underlying database.

Key characteristics:

- Has a one-to-one relationship with the database table
- Stores application data permanently
- Uses fields and data types defined in the Data Dictionary
- Can be accessed directly using Open SQL
- It is the most commonly used table type for storing business data in SAP systems.

10. What is an internal table?

An internal table is a temporary, in-memory data structure in ABAP used to store and manipulate multiple rows of data during program execution.

Key characteristics:

- Exists only while the program is running
- Stored in application server memory, not in the database
- Used for processing data such as query results before displaying or modifying it
- Can be searched, sorted, looped over, and updated dynamically
- It functions similarly to an array or collection but is designed to handle structured datasets efficiently within ABAP programs.

11. What is a work area?

A work area in ABAP is a single-row memory structure used to temporarily hold data while processing records from a database table or an internal table.

It acts like a placeholder for one record at a time, allowing you to read, modify, or transfer data before writing it back or moving to the next record. Work areas are commonly used with statements like SELECT, READ TABLE, and LOOP AT to process data sequentially.

12. What is a report in ABAP?

A report in ABAP is an executable program designed to retrieve data from the SAP database, process it according to business logic, and present the output to the user — typically in a structured list or table format.

Reports are commonly used for tasks like data analysis, business monitoring, and generating summaries, and they usually run based on user input provided through a selection screen.

13. What is a parameter vs select-option?

Parameter: A parameter is a single input field on the selection screen that accepts only one value. It is typically used when the program needs a specific, fixed input (for example, one company code or one date).

Select-Option: A select-option allows multiple values and ranges. Users can include or exclude values, define intervals (e.g., 1000–2000), and create complex selection criteria. It is internally stored as a table with fields like SIGN, OPTION, LOW, and HIGH.

14. What is a domain and a data element?

Domain: A domain defines the technical attributes of a field in the SAP Data Dictionary — such as data type, length, number of decimal places, and possible value range. It ensures consistency by enforcing uniform technical settings wherever it is used.

Data Element: A data element provides the semantic meaning of a field. It contains field labels, descriptions, and documentation that explain what the data represents from a business perspective.

15. What is the difference between START-OF-SELECTION and INITIALIZATION?

INITIALIZATION:

- Triggered when the program loads, before the selection screen is displayed.
- Mainly used to set default values, prefill parameters, or initialize variables.

START-OF-SELECTION:

- Triggered after the user executes the program (for example, presses F8).
- This is where the main processing logic typically runs — such as fetching data from the database and generating output.

16. What is a view in the Data Dictionary?

A view in the SAP Data Dictionary is a virtual table that does not store data itself but derives its data from one or more underlying database tables. It is used to combine, restrict, or present data in a specific logical format without duplicating it physically in the database, helping improve data access, security, and abstraction. Common purposes include simplifying complex queries, controlling data visibility, and providing a consistent interface for application programs.

17. Types of views in SAP?

- Database View: A database view is created by joining one or more tables using inner joins and is stored at the database level. It does not hold data itself but fetches it dynamically from underlying tables. It is mainly used for read-only data retrieval and complex selections.
- Projection View: A projection view is built on a single table and exposes only selected fields. It helps restrict unnecessary data access and simplifies data handling. It is often used to improve security and reduce interface complexity.
- Maintenance View: A maintenance view allows users to display and maintain data from multiple related tables through a single interface. It is commonly used with table maintenance generators for administrative tasks. This helps ensure data consistency while simplifying updates.
- Help View: A help view is designed to provide search help (F4 help) by combining fields from related tables. It enables users to find valid input values quickly and accurately. It improves usability by guiding users during data entry.

18. What is a selection screen?

A selection screen in ABAP is the initial input interface of a report where users enter criteria before the program executes. It allows users to filter and control the data that will be retrieved by providing inputs such as parameters and select-options, ensuring the report processes only the relevant dataset instead of scanning entire tables.

19. What is modularization in ABAP?

Modularization in ABAP is the practice of breaking a large program into smaller, independent, and reusable units to improve readability, maintainability, debugging, and overall program structure.

Common modularization techniques in ABAP include:

- Subroutines (FORM routines) — reusable code blocks within the same program.
- Function Modules — globally reusable procedures that can be called from different programs.
- Methods — functions defined inside classes as part of ABAP Objects.
- Include Programs — used to split large programs into multiple files for better organization.

20. What are function modules?

Function modules are reusable, pre-defined procedures in ABAP that encapsulate specific functionality and can be called from different programs. They are created inside a function group, have a well-defined interface (import, export, changing parameters, and tables), and are commonly used to promote modular programming, reduce code duplication, and ensure consistent business logic across the system.

21. What are subroutines?

Subroutines in ABAP are reusable blocks of code defined within a program using FORM and ENDFORM. They are called with the PERFORM statement and are mainly used to break large programs into smaller, manageable, and modular sections. Unlike function modules, subroutines are typically local to the program in which they are defined and are not designed for cross-program reuse.

22. What is macro?

A macro in ABAP is a reusable block of code defined using the DEFINE and END-OF-DEFINITION statements. Once defined, it can be invoked multiple times within the program to execute the same set of statements, helping reduce repetitive code. Macros are expanded directly at compile time, meaning the code is inserted wherever the macro is called rather than executed as a separate procedure.

23. What is smart forms?

Smart Forms is a form-building tool in SAP used to design and generate structured business documents such as invoices, purchase orders, delivery notes, and payment advices. It provides a graphical interface to define layout, text, tables, logos, and formatting, and it automatically generates the underlying function module required to print or send the document, reducing the need for complex manual coding compared to older tools like SAPscript.

24. What is ALV?

ALV (ABAP List Viewer) is a standard SAP tool used to display structured and interactive reports in a tabular format. It provides built-in features such as sorting, filtering, column reordering, totals, subtotals, exporting to Excel, and layout customization, allowing developers to create user-friendly reports without building these functionalities from scratch.

25. What are the types of ALV reports?

- Classical ALV: Generated using function modules (like REUSE_ALV_LIST_DISPLAY). Displays data in a simple list format with basic interactive features.
- Grid ALV: Uses the ALV Grid Control (CL_GUI_ALV_GRID) to present data in a more interactive table with features like editable cells, drag-and-drop columns, and better layout control.
- Object-Oriented (OO) ALV: Built using classes such as CL_SALV_TABLE. It follows object-oriented principles, requires less code, and is commonly preferred in modern ABAP development.

26. What is BAPI?

BAPI (Business Application Programming Interface) is a standardized programming interface provided by SAP SE that allows external systems or custom applications to interact with SAP business objects safely and consistently. BAPIs are pre-built function modules designed for tasks like creating sales orders, updating customer data, or posting financial transactions, ensuring data integrity while enabling integration between SAP and non-SAP systems.

27. What are enhancements in SAP?

Enhancements in SAP are techniques that allow developers to extend or customize standard SAP functionality without modifying the original source code. They are used when businesses need additional logic or features while still keeping the core system upgrade-safe, since the standard SAP code remains untouched.

Common enhancement types include:

- User Exits — predefined hooks where custom code can be inserted.
- Customer Exits — function module-based exits provided by SAP for extensions.
- BADIs (Business Add-Ins) — object-oriented enhancement spots that support multiple implementations.
- Enhancement Framework — a modern approach that includes explicit enhancement points and sections for injecting custom logic.

28. What is a user exit?

A user exit is a predefined enhancement point in standard SAP programs where developers can insert custom logic without modifying the original code. It typically involves adding code in specific includes provided by SAP, allowing organizations to tailor standard processes to their business requirements while keeping the system upgrade-safe.

29. What is a BADI?

BADI (Business Add-In) is an object-oriented enhancement technique in SAP that allows developers to add custom functionality to standard SAP applications without modifying the original code. It is based on ABAP Objects, uses interfaces and classes, and supports multiple implementations, making it more flexible and scalable than older enhancement methods like user exits while remaining upgrade-safe.

30. What is a lock object?

A lock object in ABAP is a mechanism used to control concurrent data access by preventing multiple users from modifying the same data at the same time. When a record is locked, other users can only display it but cannot make changes until the lock is released, ensuring data consistency and avoiding conflicts during database transactions.

31. What is a transport request?

A transport request in SAP is a mechanism used to move development objects — such as programs, tables, configurations, or enhancements — from one system to another within the landscape (typically Development → Quality → Production). It acts as a change management tool that records what was modified, who made the change, and ensures the objects are transferred in a controlled and trackable way, helping maintain system stability and version consistency across environments.

32. What is debugging in ABAP?

Debugging in ABAP is the process of analyzing and executing a program step by step to identify and fix errors in the code. It is done using the ABAP Debugger, which allows developers to:

- Pause program execution at breakpoints
- Monitor variable values in real time
- Trace program flow
- Modify data during execution for testing
- Identify logical or runtime errors

33. What is normalization in SAP tables?

Normalization in SAP tables is the process of organizing data into structured tables to reduce redundancy, improve data integrity, and ensure efficient storage. It involves splitting large datasets into related tables and linking them using keys (such as primary and foreign keys), so each piece of information is stored only once. This minimizes inconsistencies and makes updates more reliable across the system.

34. What is code inspector?

Code Inspector in ABAP is a static code analysis tool used to check programs for potential issues related to performance, security, syntax, and adherence to development standards. It scans the source code without executing it and helps developers identify problems such as inefficient database operations, unused variables, naming violations, and other quality risks, ensuring more reliable and optimized applications before transport or deployment.

35. What is OOABAP?

OOABAP (Object-Oriented ABAP) is the object-oriented programming model of ABAP that allows developers to build applications using classes and objects instead of relying only on procedural code.

It supports core OOP principles such as:

- Encapsulation — bundling data and methods within a class
- Inheritance — creating new classes from existing ones to reuse logic
- Polymorphism — allowing the same method to behave differently based on the object

OOABAP improves code reusability, scalability, maintainability, and aligns ABAP with modern software development practices.

36. Why did you choose SAP ABAP?

I chose SAP ABAP because it powers critical enterprise systems used by large organizations worldwide, giving me exposure to real business processes rather than just theoretical development. It offers the opportunity to build strong expertise in enterprise application development while working on high-volume, mission-critical data. Early in my career, my priority is to develop deep technical skills in a stable and widely adopted platform.

37. What have you learned so far in ABAP?

Since I recently started working with ABAP, I have focused on building a strong foundation by understanding the SAP architecture, Data Dictionary, and basic programming concepts. I've learned about internal tables, work areas, reports, selection screens, and Open SQL for database operations. Currently, I'm concentrating on strengthening these fundamentals so I can contribute effectively once assigned to a project.

38. Are you comfortable building a career in SAP?

Yes, I'm comfortable building a career in SAP because it supports mission-critical business operations for organizations globally, making it a highly relevant and stable technology. My current focus is on

developing strong enterprise-level programming skills and understanding large-scale business systems. I believe gaining deep expertise in SAP will position me well for long-term technical growth.

39. What do you know about how SAP is used in businesses?

SAP is used by businesses as an ERP platform to integrate and manage core processes such as finance, supply chain, procurement, sales, and human resources within a single system. This centralized approach enables real-time data flow across departments, improving operational efficiency and decision-making. It also helps organizations standardize processes, maintain data accuracy, and scale operations effectively.

40. What resources are you using to learn ABAP?

I'm learning ABAP through a combination of structured training and hands-on practice in the SAP system. I regularly refer to SAP documentation and internal learning materials to understand concepts correctly, and I reinforce that by experimenting with small programs and examples. This approach helps me build both conceptual clarity and practical confidence.

41. What if we put you in another domain? Are you comfortable with it?

Yes, I'm comfortable working in another domain because early in my career my priority is to build strong technical fundamentals and adapt to business requirements. While I've started with ABAP, my core strength is the ability to learn quickly and become productive in whichever technology the organization needs. I see domain changes as an opportunity to expand my skill set rather than a limitation.